

Present-Day Number Density of Extragalactic Background-Light Photons Compared with the Cosmic Microwave Background

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ABSTRACT

The cosmic microwave background (CMB) is the dominant photon population in the present universe by number density. Later astrophysical structure formation produced an additional diffuse radiation field through stellar emission, nebular emission, dust absorption and re-emission, and accretion onto black holes. This paper develops a theory-oriented framework for comparing the present-day CMB photon number density with the present-day photon number density of the optical–infrared extragalactic background light (EBL). The comparison is defined in terms of photons currently present in diffuse backgrounds, not all photons ever emitted after recombination. The CMB photon number density follows from the Planck blackbody spectrum, $n_{\gamma, \text{CMB}, 0} = [2\zeta(3)/\pi^2](k_B T_0/\hbar c)^3 \simeq 410.7 \text{ cm}^{-3}$, for the measured present-day CMB temperature. This corrects a common misframing: the number near 411 is a photon number density in cubic centimeters, not a CMB-to-astrophysical photon ratio and not the photon-to-baryon ratio. For the EBL, the photon number density is obtained from observed spectral intensity through $n_{\gamma, \text{EBL}, 0} = (4\pi/c) \int_0^\infty I_\nu(h\nu)^{-1} d\nu$. Observed optical–infrared EBL intensities imply an EBL photon density of order unity in cm^{-3} . The far-infrared and submillimeter background is numerically central: the COBE/FIRAS far-infrared background alone implies roughly $0.7\text{--}0.8 \text{ cm}^{-3}$ when converted to photon number. Adding the small optical contribution and a moderate warm-dust/mid-infrared correction gives a preferred present-day optical–infrared EBL photon density of approximately $0.7\text{--}1.2 \text{ cm}^{-3}$, with a fiducial value near $0.8\text{--}1.0 \text{ cm}^{-3}$. The corresponding CMB-to-EBL photon-number ratio is therefore approximately $3 \times 10^2\text{--}6 \times 10^2$, with a fiducial value of a few hundred. The result is a controlled present-day background comparison, not a cumulative historical count of all photons generated by baryonic processes.

Keywords: Cosmic background radiation; Extragalactic background light; Observational cosmology; Radiative transfer; Infrared background

1. INTRODUCTION

The universe contains several diffuse photon backgrounds. The most important by number density is the cosmic microwave background (CMB), a relic radiation field released when the primordial plasma became transparent near recombination. At later times, structure formation introduced additional photon sources: stars, galaxies, interstellar and intergalactic gas, dust, accreting black holes, supernovae, cosmic-ray interactions, and reionization-era sources. A natural question is therefore how the relic photon bath compares with photons produced after recombination by baryonic and astrophysical processes.

The question becomes meaningful only after the counted photon population is defined. “Photons produced by baryonic processes” may refer to photons ever

emitted by stars, photons that escaped galaxies, photons absorbed by dust, photons re-emitted in the infrared, photons currently free-streaming through intergalactic space, or photons that contribute to a measured diffuse background. These definitions are not equivalent. An ultraviolet stellar photon absorbed by dust and re-emitted as several infrared photons should not be counted both as an escaping ultraviolet photon and as a surviving infrared background photon unless the calculation is explicitly a historical production accounting. A present-day background inventory should count the photons that currently occupy the background radiation field.

This paper therefore uses the following central question:

What is the present-day number density of photons in the optical–infrared extragalactic

background light, and how does it compare with the present-day CMB photon number density?

This formulation is narrower than counting every photon ever produced after recombination, but it is physically cleaner. Both quantities are present-day physical number densities. Both are tied to observed radiation fields. The CMB side is determined by blackbody theory and the measured CMB temperature; the EBL side is determined from observed extragalactic background intensities.

A key correction concerns the number commonly quoted as approximately 411. This value is the present-day CMB photon number density in units of cm^{-3} , not a ratio of CMB photons to non-CMB photons. It is also not the photon-to-baryon ratio. Using standard cosmological parameters, the baryon number density is only of order 10^{-7} cm^{-3} , so n_γ/n_b is of order 10^9 , not 411. The corrected problem is therefore not to explain a supposed 411 : 1 CMB-to-baryonic photon ratio, but to compute and compare two present-day number densities.

The paper proceeds as follows. Section 2 defines the photon populations and scope. Section 3 reviews the relevant observational and theoretical literature. Section 4 derives the CMB photon density. Section 5 derives the conversion from EBL intensity to photon number density. Section 6 gives the numerical estimate for the present-day EBL photon density and the CMB-to-EBL photon ratio. Section 7 gives the corresponding source-term and radiative-transfer framework. Sections 8 and 9 discuss interpretation and uncertainty. Appendices provide derivations, unit checks, and source-term bookkeeping.

2. DEFINITIONS AND SCOPE

2.1. CMB photons

The CMB is treated as a nearly ideal blackbody radiation field with present temperature T_0 . Its physical photon number density is denoted $n_{\gamma, \text{CMB}}(t)$, with present value $n_{\gamma, \text{CMB}, 0}$. For freely streaming photons in an expanding universe, physical number density scales as

$$n_\gamma(t) \propto a(t)^{-3}, \quad (1)$$

where $a(t)$ is the scale factor. The corresponding comoving number density is

$$N_\gamma(t) \equiv a(t)^3 n_\gamma(t). \quad (2)$$

This comoving density is conserved if there is no photon creation or destruction. Redshift changes photon energies and frequencies, but not the number of photons in a comoving volume under free streaming.

CMB spectral distortions can in principle perturb the blackbody spectrum. Energy injection before recombination and during later epochs may produce μ -type, y -type,

or intermediate distortions. For the present comparison, these corrections are negligible. Processes such as the thermal Sunyaev–Zeldovich effect redistribute photon energies and affect anisotropies or spectral shape, but they do not significantly alter the total CMB photon count at the order relevant here. The main calculation therefore uses the ideal blackbody photon density [Kogut et al. \(2011\)](#); [Chluba et al. \(2012\)](#).

2.2. Extragalactic background-light photons

The EBL is the diffuse extragalactic radiation field generated by stars, galaxies, dust, and accreting black holes over cosmic history. In this paper, EBL refers primarily to radiation from the ultraviolet through the optical, near-infrared, mid-infrared, far-infrared, and sub-millimeter, roughly the standard optical–infrared EBL domain. It is commonly separated into the cosmic optical background (COB) and cosmic infrared background (CIB). The COB is dominated by direct starlight and nebular emission that escaped galaxies, while the CIB is dominated by starlight absorbed by dust and re-emitted at longer wavelengths [Hauser & Dwek \(2001\)](#); [Dwek & Krennrich \(2013\)](#); [Cooray \(2016\)](#).

The main comparison excludes the CMB, local Solar System foregrounds, Galactic emission, and photons that were emitted but absorbed before joining a freely streaming extragalactic background. It also excludes radio, X-ray, and gamma-ray backgrounds from the primary numerical estimate. Those backgrounds are physically important, but they involve different measurement systematics and do not dominate the optical–infrared EBL photon budget. This paper compares the CMB with the present-day optical–infrared EBL, not with every post-recombination photon ever emitted.

The present-day comparison ratio is defined as

$$R_{\text{EBL/CMB}, 0} \equiv \frac{n_{\gamma, \text{EBL}, 0}}{n_{\gamma, \text{CMB}, 0}}, \quad (3)$$

or, equivalently,

$$R_{\text{CMB/EBL}, 0} \equiv \frac{n_{\gamma, \text{CMB}, 0}}{n_{\gamma, \text{EBL}, 0}}. \quad (4)$$

2.3. Physical density, comoving density, emissivity, and production rate density

The distinction between physical density, comoving density, emissivity, and production rate density is central. A physical photon number density has units such as cm^{-3} or m^{-3} . A comoving number density is a physical density multiplied by a^3 , with units per comoving volume. A photon production rate density has units such as photons $\text{s}^{-1} \text{ Mpc}^{-3}$. These quantities are not interchangeable.

A schematic physical-density evolution equation is

$$\frac{dn_\gamma}{dt} + 3Hn_\gamma = \dot{n}_{\gamma,\text{src}} - \dot{n}_{\gamma,\text{sink}}, \quad (5)$$

where H is the Hubble parameter, $\dot{n}_{\gamma,\text{src}}$ represents photon injection into the counted population, and $\dot{n}_{\gamma,\text{sink}}$ represents removal from that population. In a realistic EBL calculation, these terms are spectral and redshift-dependent rather than single bolometric numbers.

In the radiative-transfer expressions below, $\epsilon_\nu(\nu, z)$ denotes a *comoving* spectral emissivity: radiated energy per unit proper time, per unit emitted frequency, per unit comoving volume. Its SI units are $\text{W Hz}^{-1} \text{m}_{\text{comoving}}^{-3}$, or equivalently $\text{J s}^{-1} \text{Hz}^{-1} \text{m}_{\text{comoving}}^{-3}$. If instead one uses a proper spectral emissivity, it differs by the volume factor

$$\epsilon_\nu^{\text{proper}}(\nu, z) = (1+z)^3 \epsilon_\nu^{\text{comoving}}(\nu, z), \quad (6)$$

and the redshift factors in the observed-intensity integral must be changed accordingly. This paper uses the comoving convention throughout.

3. BACKGROUND AND LITERATURE REVIEW

The CMB temperature and blackbody spectrum are among the best measured quantities in cosmology. COBE/FIRAS established that the CMB spectrum is extremely close to a blackbody [Mather et al. \(1994\)](#); [Fixsen et al. \(1996\)](#), and Fixsen’s temperature determination gives the standard monopole temperature used for the numerical photon-density calculation [Fixsen \(2009\)](#). The blackbody formula then fixes the photon number density without requiring a model of late-time astrophysical sources.

The EBL literature is broader because the signal is faint and strongly foreground-limited. Reviews by [Hauser and Dwek](#), [Dwek and Krennrich](#), and [Cooray](#) define the EBL as the accumulated optical–infrared light of stars, galaxies, dust, and accreting black holes, excluding the CMB [Hauser & Dwek \(2001\)](#); [Dwek & Krennrich \(2013\)](#); [Cooray \(2016\)](#). These reviews also emphasize that the COB and CIB must be treated together: direct starlight dominates the optical and near-infrared background, while dust-reprocessed radiation dominates the mid-infrared through submillimeter background.

Galaxy-count measurements provide robust lower limits on the EBL. [Driver et al.](#) measured the integrated light from the far ultraviolet to far infrared using deep ground- and space-based galaxy counts, finding comparable optical and infrared energy contributions but with the infrared somewhat larger in their accounting [Driver et al. \(2016\)](#). [Saldana-Lopez et al.](#) constructed an observationally based evolving EBL model, obtaining a local bolometric optical–infrared EBL intensity near

$55 \text{ nW m}^{-2} \text{sr}^{-1}$, split into a COB near $23 \text{ nW m}^{-2} \text{sr}^{-1}$ and a CIB near $32 \text{ nW m}^{-2} \text{sr}^{-1}$ [Saldana-Lopez et al. \(2021\)](#). These works are especially important because they tie the EBL estimate directly to resolved or reconstructed galaxy populations.

Direct absolute EBL measurements are more difficult. Zodiacal light is a severe foreground in the optical and near-infrared, while Galactic dust is a severe foreground in the far-infrared and submillimeter. Near-infrared background-anisotropy measurements provide a complementary probe of unresolved structure, although they are not the primary bolometric anchor for the present photon-count estimate [Zemcov et al. \(2014\)](#). COBE/DIRBE and FIRAS measurements provided foundational infrared background constraints [Puget et al. \(1996\)](#); [Hauser et al. \(1998\)](#); [Fixsen et al. \(1998\)](#). The COBE/FIRAS far-infrared background measurement is particularly important for photon counting because it directly anchors the low-energy tail where each unit of radiative energy corresponds to many photons. Recent optical-background measurements using New Horizons illustrate both the value and the difficulty of direct measurements beyond the bright inner Solar System foregrounds [Symons et al. \(2023\)](#); [Postman et al. \(2024\)](#).

Gamma-ray attenuation gives an independent EBL constraint. Very high energy gamma rays from blazars can pair-produce with EBL photons, so their observed spectra carry information about the intervening optical–infrared photon field. This method has been used by Fermi-LAT, H.E.S.S., and related analyses [Ackermann et al. \(2012\)](#); [H.E.S.S. Collaboration \(2013\)](#); [Desai et al. \(2019\)](#); [Aharonian et al. \(2006\)](#). These constraints generally support an EBL close to galaxy-count lower limits, although they depend on assumptions about intrinsic gamma-ray source spectra.

The source mix is also established by modeling. EBL models and lower-limit reconstructions by [Finke et al.](#), [Dominguez et al.](#), [Gilmore et al.](#), [Franceschini et al.](#), and [Kneiske and Dole](#) show that star-forming galaxies dominate the optical–infrared EBL energy budget, with AGN contributing but not dominating [Finke et al. \(2010\)](#); [Dominguez et al. \(2011\)](#); [Gilmore et al. \(2012\)](#); [Franceschini et al. \(2008\)](#); [Kneiske & Dole \(2010\)](#). [Dominguez et al.](#) give a fiducial low-redshift source decomposition in which AGN-like systems are a modest bolometric contribution, while star-forming and starburst populations supply most of the optical–infrared EBL [Dominguez et al. \(2011\)](#). Cosmic star-formation history measurements and reviews provide the time-dependent backbone for theoretical EBL source models [Hopkins & Beacom \(2006\)](#); [Madau & Dickinson \(2014\)](#).

4. CMB PHOTON NUMBER DENSITY

The CMB photon number density follows from the Planck blackbody spectrum, using standard physical constants [Mohr et al. \(2025\)](#). The blackbody spectral energy density is

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{\exp(h\nu/k_B T) - 1}, \quad (7)$$

with units $\text{J m}^{-3} \text{Hz}^{-1}$. Dividing by the photon energy $h\nu$ gives the spectral photon number density,

$$\frac{dn_\gamma}{d\nu} = \frac{u_\nu}{h\nu} = \frac{8\pi\nu^2}{c^3} \frac{1}{\exp(h\nu/k_B T) - 1}. \quad (8)$$

Integrating over frequency gives

$$n_\gamma(T) = \int_0^\infty \frac{8\pi\nu^2}{c^3} \frac{1}{\exp(h\nu/k_B T) - 1} d\nu. \quad (9)$$

With $x = h\nu/k_B T$, this becomes

$$n_\gamma(T) = \frac{8\pi}{c^3} \left(\frac{k_B T}{h} \right)^3 \int_0^\infty \frac{x^2}{e^x - 1} dx. \quad (10)$$

Using

$$\int_0^\infty \frac{x^2}{e^x - 1} dx = 2\zeta(3), \quad (11)$$

one obtains

$$n_\gamma(T) = \frac{16\pi\zeta(3)}{c^3} \left(\frac{k_B T}{h} \right)^3 = \frac{2\zeta(3)}{\pi^2} \left(\frac{k_B T}{hc} \right)^3. \quad (12)$$

Using $T_0 \simeq 2.72548 \text{ K}$, the present-day CMB photon density is

$$n_{\gamma, \text{CMB}, 0} \simeq 410.7 \text{ cm}^{-3}. \quad (13)$$

The fractional uncertainty is negligible for the present comparison because $\delta n_\gamma/n_\gamma = 3\delta T/T$. A temperature uncertainty of order $6 \times 10^{-4} \text{ K}$ therefore changes $n_{\gamma, \text{CMB}, 0}$ only at the 10^{-4} fractional level.

This value is the source of the commonly cited number near 411. It is a number density, not a ratio. If the present baryon density is estimated from $\Omega_b \rho_c$, the baryon number density is about $2.5 \times 10^{-7} \text{ cm}^{-3}$, so the photon-to-baryon ratio is of order 1.6×10^9 , while n_b/n_γ is of order 6×10^{-10} [Planck Collaboration \(2020\)](#). Those ratios are different quantities from $n_{\gamma, \text{CMB}, 0}$.

5. FROM EBL INTENSITY TO PHOTON NUMBER DENSITY

The EBL is not thermal. Its photon number density must be inferred from the observed spectral intensity. For an isotropic background with specific intensity I_ν , the spectral energy density is

$$u_\nu = \frac{4\pi}{c} I_\nu. \quad (14)$$

The spectral photon number density is therefore

$$\frac{dn_\gamma}{d\nu} = \frac{u_\nu}{h\nu} = \frac{4\pi}{c} \frac{I_\nu}{h\nu}. \quad (15)$$

Thus,

$$n_{\gamma, \text{EBL}, 0} = \frac{4\pi}{c} \int_0^\infty \frac{I_\nu}{h\nu} d\nu. \quad (16)$$

If the EBL spectrum is tabulated by wavelength, the equivalent expression is

$$n_{\gamma, \text{EBL}, 0} = \frac{4\pi}{hc^2} \int_0^\infty \lambda I_\lambda d\lambda. \quad (17)$$

Equations in wavelength space require λ and I_λ to be in consistent units; for the SI expression above, λ is measured in meters and I_λ in $\text{W m}^{-2} \text{sr}^{-1} \text{m}^{-1}$.

Many EBL plots are shown as λI_λ versus λ . In that case,

$$n_{\gamma, \text{EBL}, 0} = \frac{4\pi}{hc^2} \int_0^\infty (\lambda I_\lambda) d\lambda. \quad (18)$$

If the horizontal axis is logarithmic, then $d\lambda = \lambda d \ln \lambda$, so the photon-counting weight differs from the bolometric energy integral. These expressions show why photon number density and bolometric energy intensity are not equivalent. A fixed radiative energy density corresponds to more photons when the photons have lower energy. Therefore the far-infrared and submillimeter parts of the EBL are especially important for photon number density, even when the optical and infrared backgrounds have comparable total energy intensities.

6. NUMERICAL ESTIMATE OF THE PRESENT-DAY EBL PHOTON DENSITY

6.1. Energy-density scale

A typical bolometric optical–infrared EBL intensity is of order several tens of $\text{nW m}^{-2} \text{sr}^{-1}$. [Driver et al.](#) and [Saldana-Lopez et al.](#) support a local optical–infrared total near $50\text{--}55 \text{ nW m}^{-2} \text{sr}^{-1}$, depending on whether one emphasizes galaxy-count lower limits or reconstructed EBL models [Driver et al. \(2016\)](#); [Saldana-Lopez et al. \(2021\)](#). As a round fiducial scale, take

$$I_{\text{EBL}} \sim 50 \text{ nW m}^{-2} \text{sr}^{-1}. \quad (19)$$

The corresponding radiation energy density is

$$u_{\text{EBL}} = \frac{4\pi}{c} I_{\text{EBL}} \simeq 2.1 \times 10^{-15} \text{ J m}^{-3}. \quad (20)$$

For comparison, the CMB energy density is much larger, but the contrast in photon number is even stronger because the characteristic CMB photon energy is much smaller than the characteristic EBL photon energy.

6.2. Optical and near-infrared component

For the cosmic optical background, take an observationally motivated intensity of roughly

$$I_{\text{COB}} \simeq 23\text{--}28 \text{ nW m}^{-2} \text{ sr}^{-1}, \quad (21)$$

consistent with galaxy-count and gamma-ray-attenuation based low-EBL estimates [Driver et al. \(2016\)](#); [Saldana-Lopez et al. \(2021\)](#); [Desai et al. \(2019\)](#). The energy density associated with $I_{\text{COB}} = 23 \text{ nW m}^{-2} \text{ sr}^{-1}$ is

$$u_{\text{COB}} = \frac{4\pi}{c} I_{\text{COB}} \simeq 9.6 \times 10^{-16} \text{ J m}^{-3}. \quad (22)$$

For photons with characteristic energy near 1–1.2 eV, this gives

$$n_{\gamma, \text{COB}, 0} \sim (5\text{--}6) \times 10^{-3} \text{ cm}^{-3}. \quad (23)$$

Thus the direct optical and near-infrared background contributes only a small number density compared with both the CMB and the infrared EBL. Direct optical-background uncertainties matter for the optical energy budget, but they barely move the total EBL photon count because the count is dominated by lower-energy infrared photons.

6.3. Far-infrared and submillimeter anchor

The far-infrared/submillimeter background should be the quantitative anchor for the EBL photon number density. [Fixsen et al.](#) measured the extragalactic far-infrared background from COBE/FIRAS over approximately 125–2000 μm , with a total intensity near

$$I_{\text{FIRB}} \simeq 14 \text{ nW m}^{-2} \text{ sr}^{-1} \quad (24)$$

and a modified-blackbody spectral shape of the form

$$I_{\nu} \propto \nu^{0.64} B_{\nu}(18.5 \text{ K}) \quad (25)$$

within the fitted far-infrared domain [Fixsen et al. \(1998\)](#). Converting this measured low-energy component to photon number gives a representative photon energy of only a few 10^{-3} eV, so

$$n_{\gamma, \text{FIRB}, 0} \sim 0.7\text{--}0.8 \text{ cm}^{-3}. \quad (26)$$

This single empirical component already contains almost one photon per cubic centimeter. It therefore rules out an EBL photon-density lower bound as small as 0.3 cm^{-3} for the full optical–infrared EBL, because the far-infrared/submillimeter background alone is larger than that.

The physical reason is simple. For a given intensity, photon number scales approximately as inverse photon energy:

$$n_{\gamma} \sim \frac{4\pi I/c}{\bar{E}_{\gamma}}. \quad (27)$$

The far-infrared background carries less energy than the full EBL, but its photons have energies roughly two orders of magnitude below optical photons. Consequently it dominates the photon count.

6.4. Warm-dust and mid-infrared correction

The total CIB is larger than the FIRAS far-infrared/submillimeter band alone. [Saldana-Lopez et al.](#) give a local CIB intensity near

$$I_{\text{CIB}} \simeq 32 \text{ nW m}^{-2} \text{ sr}^{-1}, \quad (28)$$

so after the $\sim 14 \text{ nW m}^{-2} \text{ sr}^{-1}$ FIR/submillimeter component there remains an order $18 \text{ nW m}^{-2} \text{ sr}^{-1}$ mid-infrared and warmer-dust contribution [Saldana-Lopez et al. \(2021\)](#); [Fixsen et al. \(1998\)](#). This residual emission has higher characteristic photon energies than the cold FIR/submillimeter tail, but lower energies than optical starlight. EBL dust models using cold, warm, and hot components therefore place this correction at approximately

$$n_{\gamma, \text{midIR}, 0} \sim 0.05\text{--}0.4 \text{ cm}^{-3}, \quad (29)$$

with the most plausible contribution narrower than the full envelope because the residual cannot all be assigned to the coldest dust component [Finke et al. \(2010\)](#). The correction is non-negligible, but it does not replace the FIR/submillimeter anchor as the dominant term.

6.5. Combined EBL photon density

Putting the pieces together gives

$$n_{\gamma, \text{EBL}, 0} \simeq n_{\gamma, \text{FIRB}, 0} + n_{\gamma, \text{midIR}, 0} + n_{\gamma, \text{COB}, 0}. \quad (30)$$

Numerically,

$$n_{\gamma, \text{EBL}, 0} \approx 0.7\text{--}1.2 \text{ cm}^{-3}, \quad (31)$$

with a fiducial value near

$$n_{\gamma, \text{EBL}, 0} \sim 0.8\text{--}1.0 \text{ cm}^{-3}. \quad (32)$$

A deliberately conservative envelope of $0.5\text{--}1.5 \text{ cm}^{-3}$ would still be defensible for a broad theory discussion, but the better literature-anchored estimate is closer to $0.7\text{--}1.2 \text{ cm}^{-3}$. The older broad range $0.3\text{--}3 \text{ cm}^{-3}$ is too loose: the lower edge is already below the measured far-infrared/submillimeter contribution alone, while the upper edge would require assigning too much of the EBL energy to implausibly low mean photon energies.

6.6. Ratio to the CMB

Using $n_{\gamma, \text{CMB}, 0} \simeq 410.7 \text{ cm}^{-3}$, the preferred EBL range gives

$$\frac{n_{\gamma, \text{EBL}, 0}}{n_{\gamma, \text{CMB}, 0}} \approx \frac{0.7-1.2}{410.7} \simeq (1.7-2.9) \times 10^{-3}. \quad (33)$$

The inverse ratio is

$$\frac{n_{\gamma, \text{CMB}, 0}}{n_{\gamma, \text{EBL}, 0}} \approx 3.4 \times 10^2 - 5.9 \times 10^2. \quad (34)$$

Thus the CMB outnumbers the present optical-infrared EBL by a factor of a few hundred. The exact value depends mainly on the adopted far-infrared and mid-infrared spectral shape, not on the optical background.

7. SOURCE-TERM THEORY OF THE EBL

7.1. General radiative-transfer form

The empirical estimate above is based on observed background intensity. A complementary theory framework describes how the EBL arises from cosmic emissivity. With the comoving spectral-emissivity convention defined in Section 2.3, the observed specific intensity at present-day observer frequency ν_0 can be written schematically as

$$I_{\nu_0} = \frac{c}{4\pi} \int_0^\infty \frac{\epsilon_\nu[(1+z)\nu_0, z]}{H(z)(1+z)} e^{-\tau(\nu_0, z)} dz, \quad (35)$$

where $\nu = (1+z)\nu_0$ is the emitted frequency corresponding to observed frequency ν_0 , ϵ_ν is the comoving spectral emissivity, and τ is an effective optical depth. This is the present-day observer form, with $z_0 = 0$. Once I_{ν_0} is known, the photon density follows from

$$n_{\gamma, 0} = \frac{4\pi}{c} \int_0^\infty \frac{I_{\nu_0}}{h\nu_0} d\nu_0. \quad (36)$$

This form is preferable to a naive sum of photon source terms because it keeps track of spectral shape, redshift, emissivity evolution, and absorption. It also avoids double-counting dust-reprocessed radiation.

If a proper spectral emissivity is used instead, $\epsilon_\nu^{\text{proper}} = (1+z)^3 \epsilon_\nu^{\text{comoving}}$. Substitution into the same present-day intensity expression changes the redshift weighting of the integrand. The displayed equation is therefore not convention-free; it is correct under the comoving-emissivity convention stated above.

7.2. Cosmic star formation and stellar emissivity

The dominant energy source for the optical-infrared EBL is stellar radiation integrated over cosmic history. The cosmic star-formation rate density, usually denoted $\psi(z)$, rises from early times to a broad maximum around $z \sim 2$ and then declines toward the present [Hopkins &](#)

[Beacom \(2006\)](#); [Madau & Dickinson \(2014\)](#). A schematic intrinsic stellar emissivity may be written as

$$\epsilon_{\nu,*}^{\text{int}}(\nu, z) = \int_0^{t(z)} \psi[t(z) - t'] \ell_\nu(\nu, t', Z) dt', \quad (37)$$

where ℓ_ν is the spectral luminosity per unit stellar mass of a stellar population of age t' and metallicity Z . In practice, EBL models compute this using stellar population synthesis and galaxy-evolution constraints.

Only a fraction of intrinsic stellar radiation escapes directly into the extragalactic background. If $f_{\text{abs}}(\nu, z)$ is the fraction absorbed by dust, the escaped stellar emissivity is

$$\epsilon_{\nu,*}^{\text{esc}}(\nu, z) = [1 - f_{\text{abs}}(\nu, z)] \epsilon_{\nu,*}^{\text{int}}(\nu, z). \quad (38)$$

This term primarily builds the COB.

7.3. Dust absorption and infrared re-emission

Dust absorbs part of the ultraviolet and optical stellar radiation and re-emits the energy thermally in the infrared. The absorbed stellar energy production rate density is schematically

$$\epsilon_{\text{abs},*}(z) = \int_0^\infty f_{\text{abs}}(\nu, z) \epsilon_{\nu,*}^{\text{int}}(\nu, z) d\nu. \quad (39)$$

If $\phi_{\nu, \text{dust}}(\nu, z)$ is a normalized dust emission spectrum satisfying

$$\int_0^\infty \phi_{\nu, \text{dust}}(\nu, z) d\nu = 1, \quad (40)$$

then the dust emissivity is

$$\epsilon_{\nu, \text{dust}}(\nu, z) = \epsilon_{\text{abs},*}(z) \phi_{\nu, \text{dust}}(\nu, z). \quad (41)$$

This term primarily builds the CIB.

Dust reprocessing is especially important for photon number. A single optical or ultraviolet photon can be absorbed and re-emitted as multiple lower-energy infrared photons. Energy is approximately conserved, but photon number is not. Therefore a photon-counting calculation must not count both the absorbed stellar photon and the re-emitted infrared photons as separate surviving background photons. The escaped background contains the direct photons that escaped plus the reprocessed photons that escaped, not the absorbed intermediate photons.

7.4. AGN emission

Accretion onto supermassive black holes contributes to the EBL through direct optical, ultraviolet, X-ray, and infrared emission, as well as dust-reprocessed radiation from obscuring structures. A schematic AGN contribution may be written

$$\epsilon_{\nu, \text{AGN}}^{\text{esc}}(\nu, z) = [1 - f_{\text{abs}, \text{AGN}}(\nu, z)] \epsilon_{\nu, \text{AGN}}^{\text{int}}(\nu, z), \quad (42)$$

with an additional infrared dust term for absorbed AGN radiation. AGN are important in some bands but are generally subdominant in the bolometric optical–infrared EBL compared with star-forming galaxies [Dominguez et al. \(2011\)](#); [Saldana-Lopez et al. \(2021\)](#). For the present photon-density estimate, AGN are therefore best treated as a secondary correction rather than as a leading source term.

7.5. *Minor and non-leading photon sources*

Supernovae, cosmic-ray interactions, and reionization-era sources are not leading terms in the present optical–infrared EBL photon number density. Supernovae produce optical, ultraviolet, X-ray, and gamma-ray photons, but their integrated optical–infrared contribution is small compared with ordinary stellar and dust emission. Cosmic-ray interactions are more relevant to gamma-ray and high-energy backgrounds than to the optical–infrared EBL count. Reionization-era photons are physically crucial because they changed the ionization state of the intergalactic medium, but ionizing photons at $z \sim 6$ –15 do not simply accumulate as a large surviving optical–infrared background. Many are absorbed by neutral hydrogen or processed into other radiation. Reionization-era near-infrared backgrounds are therefore too small to affect the present optical–infrared photon inventory at the order considered here [Cooray et al. \(2012\)](#); [Robertson et al. \(2015\)](#); [Haardt & Madau \(2012\)](#).

These processes can be included in a comprehensive source inventory, but they should not set the main numerical estimate. The corrected source expression for the optical–infrared EBL is more compactly written as

$$\epsilon_{\nu}^{\text{EBL}} = \epsilon_{\nu,*}^{\text{esc}} + \epsilon_{\nu,\text{dust}} + \epsilon_{\nu,\text{AGN}}^{\text{esc}} + \epsilon_{\nu,\text{AGN,dust}} + \epsilon_{\nu,\text{minor}}, \quad (43)$$

where $\epsilon_{\nu,\text{minor}}$ contains small terms only when they are relevant to the wavelength range being modeled. This spectral emissivity is inserted into the radiative-transfer equation, and the resulting intensity is converted to photon density. The primary object is not a loose cumulative photon sum, but an emissivity model linked to the observed background field.

8. DISCUSSION

The central physical result is that the CMB dominates the present photon number density by a large factor. The CMB contains approximately $410.7 \text{ photons cm}^{-3}$. The optical–infrared EBL contains roughly one photon per cubic centimeter, more specifically about 0.7 – 1.2 cm^{-3} in a low-EBL observational picture. Thus, the CMB outnumbers the EBL by a factor of a few hundred.

This does not mean the EBL is unimportant. The EBL records the integrated radiative history of galaxy

formation, star formation, dust processing, and black-hole accretion. It is a major observational probe of galaxy evolution and high-energy gamma-ray opacity. The reason it is small by photon number is that its photons are much more energetic than CMB photons. A typical CMB photon has an energy of order 10^{-4} eV , while EBL photons range from a few 10^{-3} eV in the far-infrared/submillimeter to 1 eV or more in the optical.

The comparison also clarifies the difference between energy density and number density. The EBL is much closer to the CMB in total radiative energy than in photon count because infrared dust processing lowers photon energies and thereby raises photon counts, but the CMB still wins by hundreds in total number density. Two radiation backgrounds can have energy densities within a few orders of magnitude while having photon number densities that differ by much more, because photon number density is approximately energy density divided by mean photon energy. The low photon energy of the CMB is precisely why its number density is so large.

Dust is crucial in the EBL photon count. Dust-reprocessed radiation shifts energy from optical and ultraviolet wavelengths into the infrared. For a fixed energy budget, this increases the number of photons. Therefore the CIB, and especially its far-infrared/submillimeter tail, dominates the EBL photon number even when the COB and CIB are comparable within a factor of order unity in bolometric intensity. This is also why double-counting must be avoided: absorbed stellar photons and emitted dust photons are related stages in the same radiative energy flow.

A fully time-dependent comparison of the EBL and CMB photon densities would require an evolving EBL spectral model, including redshift-dependent star formation, metallicity, dust opacity, AGN evolution, and intergalactic absorption. Such a calculation is feasible within established EBL modeling frameworks, but it is a different project from a present-day background comparison. The present-day calculation is cleaner because it is tied directly to observed background intensities.

9. SYSTEMATIC UNCERTAINTIES AND SCOPE CONDITIONS

The CMB side of the calculation has very small uncertainty for the present purpose. The CMB monopole temperature is well measured, and the blackbody formula fixes the photon density. Spectral distortions are important for precision cosmology but negligible for the order-of-magnitude CMB-to-EBL comparison.

The EBL side has larger uncertainties. Three observational tiers should be distinguished. First, galaxy counts

provide robust lower limits because resolved galaxies must contribute at least their summed light to the EBL. Counts may still miss diffuse low-surface-brightness emission, faint galaxies, or intrahalo light. Second, gamma-ray attenuation reconstructions provide an independent constraint because high-energy gamma rays interact with EBL photons through pair production, but they depend on assumptions about intrinsic source spectra and propagation. Third, direct absolute measurements can in principle capture all components, but they are highly sensitive to foreground modeling.

The foreground problem is severe. Zodiacal light from interplanetary dust is a major contaminant in the optical and near-infrared. Galactic stars and diffuse Galactic light must also be subtracted. In the far-infrared and submillimeter, Galactic dust emission is a major foreground. These systematics explain why galaxy-count lower limits, direct photometric measurements, and gamma-ray attenuation constraints can differ. Recent New Horizons optical measurements are valuable because they reduce zodiacal-light contamination, but they still illustrate tension between direct COB estimates and galaxy-count expectations [Symons et al. \(2023\)](#); [Postman et al. \(2024\)](#).

For the present photon-density problem, however, optical disagreements are less important than they might appear. The optical contribution to the photon count is only of order 10^{-3} – 10^{-2} cm^{-3} , while the FIR/submillimeter contribution is already of order 0.7 – 0.8 cm^{-3} . Thus, direct optical-background uncertainty matters more for the EBL energy budget than for the total EBL photon number density.

The conversion from EBL intensity to photon number density adds another uncertainty. Photon number weights low-energy photons strongly. Therefore uncertainties in the far-infrared and submillimeter tail have more leverage on $n_{\gamma,\text{EBL}}$ than comparable fractional uncertainties in the optical background. This is why the final EBL photon number density is best stated as a literature-anchored estimate, approximately 0.7 – 1.2 cm^{-3} , rather than as a precision measurement.

The final scope condition is definitional. This paper compares the CMB with the optical–infrared EBL, not with every non-CMB photon in the universe. A broader inventory could include radio, X-ray, gamma-ray, and local radiation fields, but that would no longer be the standard optical–infrared EBL problem. It would also require different observational inputs and source modeling.

10. CONCLUSION

The present-day CMB photon density is

$$n_{\gamma,\text{CMB},0} \simeq 410.7 \text{ cm}^{-3}. \quad (44)$$

This value follows directly from the Planck blackbody spectrum and the measured CMB temperature. It is the correct interpretation of the commonly cited number near 411: it is a physical photon number density, not a ratio.

The present-day optical–infrared EBL photon density is much smaller but still of order unity:

$$n_{\gamma,\text{EBL},0} \approx 0.7\text{--}1.2 \text{ cm}^{-3}, \quad (45)$$

with a fiducial value near 0.8 – 1.0 cm^{-3} . The far-infrared/submillimeter background is the empirical backbone of this estimate, contributing roughly 0.7 – 0.8 cm^{-3} by itself.

The resulting comparison is

$$\frac{n_{\gamma,\text{CMB},0}}{n_{\gamma,\text{EBL},0}} \approx 3 \times 10^2\text{--}6 \times 10^2, \quad (46)$$

with a fiducial value of a few hundred. The result is physically intuitive. The CMB is a cold relic blackbody with extremely many low-energy photons. The EBL is the integrated optical–infrared radiation field produced by astrophysical structure formation, but its photons are fewer because they are typically much more energetic. A careful comparison must therefore be framed in terms of present-day background photon number densities, not a cumulative sum of all photons ever produced by baryonic processes.

APPENDIX

A. APPENDIX A. DERIVATION OF THE CMB PHOTON NUMBER DENSITY

The Planck spectral energy density is

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}. \quad (\text{A1})$$

The photon energy is $E = h\nu$, so

$$dn_\gamma = \frac{u_\nu d\nu}{h\nu}. \quad (\text{A2})$$

Therefore,

$$n_\gamma(T) = \int_0^\infty \frac{8\pi\nu^2}{c^3} \frac{1}{e^{h\nu/k_B T} - 1} d\nu. \quad (\text{A3})$$

Let

$$x = \frac{h\nu}{k_B T}. \quad (\text{A4})$$

Then

$$\nu = \frac{k_B T}{h} x, \quad d\nu = \frac{k_B T}{h} dx. \quad (\text{A5})$$

Substitution gives

$$n_\gamma(T) = \frac{8\pi}{c^3} \left(\frac{k_B T}{h} \right)^3 \int_0^\infty \frac{x^2}{e^x - 1} dx. \quad (\text{A6})$$

Using the expansion

$$\frac{1}{e^x - 1} = \sum_{m=1}^\infty e^{-mx}, \quad (\text{A7})$$

one finds

$$\int_0^\infty \frac{x^2}{e^x - 1} dx = \sum_{m=1}^\infty \int_0^\infty x^2 e^{-mx} dx = \sum_{m=1}^\infty \frac{2}{m^3} = 2\zeta(3). \quad (\text{A8})$$

Therefore,

$$n_\gamma(T) = \frac{16\pi\zeta(3)}{c^3} \left(\frac{k_B T}{h} \right)^3. \quad (\text{A9})$$

Since $h = 2\pi\hbar$, this is equivalent to

$$n_\gamma(T) = \frac{2\zeta(3)}{\pi^2} \left(\frac{k_B T}{\hbar c} \right)^3. \quad (\text{A10})$$

The unit check is

$$\left(\frac{k_B T}{h} \right)^3 \frac{1}{c^3} \sim \frac{s^{-3}}{m^3 s^{-3}} = m^{-3}. \quad (\text{A11})$$

B. APPENDIX B. CONVERSION FROM EBL INTENSITY TO PHOTON NUMBER DENSITY

For an isotropic radiation field, the spectral energy density and specific intensity are related by

$$u_\nu = \frac{4\pi}{c} I_\nu. \quad (\text{B12})$$

The spectral photon number density is

$$\frac{dn_\gamma}{d\nu} = \frac{u_\nu}{h\nu} = \frac{4\pi}{c} \frac{I_\nu}{h\nu}. \quad (\text{B13})$$

Integrating gives

$$n_\gamma = \frac{4\pi}{c} \int_0^\infty \frac{I_\nu}{h\nu} d\nu. \quad (\text{B14})$$

To convert to wavelength space, use

$$\nu = \frac{c}{\lambda}, \quad h\nu = \frac{hc}{\lambda}. \quad (\text{B15})$$

The bolometric intensity satisfies $I_\nu d\nu = I_\lambda d\lambda$ up to the sign associated with $d\nu/d\lambda < 0$; the integration limits absorb the sign. Therefore,

$$n_\gamma = \frac{4\pi}{c} \int_0^\infty \frac{I_\lambda}{hc/\lambda} d\lambda = \frac{4\pi}{hc^2} \int_0^\infty \lambda I_\lambda d\lambda. \quad (\text{B16})$$

This is the wavelength-space expression used for EBL spectra. If I_λ is not expressed per meter, the wavelength unit conversion must be applied explicitly.

C. APPENDIX C. COSMOLOGICAL RADIATIVE TRANSFER AND REDSHIFT DEPENDENCE

The observed EBL intensity can be related to source emissivity through cosmological radiative transfer. In this paper $\epsilon_\nu(\nu, z)$ is a comoving spectral emissivity at emitted frequency ν , with units of energy per proper time, per emitted-frequency interval, per comoving volume. Photons observed at frequency ν_0 were emitted at

$$\nu = (1 + z)\nu_0. \quad (\text{C17})$$

A standard present-day observer form for the observed specific intensity is

$$I_{\nu_0} = \frac{c}{4\pi} \int_0^\infty \frac{\epsilon_\nu[(1+z)\nu_0, z]}{H(z)(1+z)} e^{-\tau(\nu_0, z)} dz. \quad (\text{C18})$$

The factor $H(z)^{-1}$ converts redshift interval into a propagation time or distance scale. The factor $(1+z)^{-1}$ accounts for cosmological redshifting and time dilation in the observed intensity under the comoving-emissivity convention. The attenuation factor $e^{-\tau}$ represents absorption between emission and observation.

If one instead defines a proper spectral emissivity, then

$$\epsilon_\nu^{\text{proper}}(\nu, z) = (1+z)^3 \epsilon_\nu^{\text{comoving}}(\nu, z), \quad (\text{C19})$$

and the equivalent present-day integral must be rewritten with the corresponding redshift factor. The physical photon number density associated with the observed background is then

$$n_{\gamma, 0} = \frac{4\pi}{c} \int_0^\infty \frac{I_{\nu_0}}{h\nu_0} d\nu_0. \quad (\text{C20})$$

This formula connects source evolution to the present-day photon number density.

D. APPENDIX D. SOURCE-TERM UNITS AND BOOKKEEPING

A photon production rate density has units

$$[\dot{N}_\gamma] = \text{photons s}^{-1} \text{Mpc}^{-3} \quad (\text{D21})$$

if it is comoving. Integrating over proper time gives a comoving photon number density,

$$N_\gamma(t) = \int_{t_i}^t \dot{N}_{\gamma, \text{src}}(t') dt', \quad (\text{D22})$$

only if photon destruction, absorption, and reprocessing are absent or explicitly included. The corresponding physical density is

$$n_\gamma(t) = \frac{N_\gamma(t)}{a(t)^3}. \quad (\text{D23})$$

For stellar emission, a schematic photon production rate density can be written

$$\dot{N}_{\gamma, *}(t) = \psi(t) Y_{\gamma, *}(t), \quad (\text{D24})$$

where $\psi(t)$ is the cosmic star-formation rate density and $Y_{\gamma, *}$ is the photon yield per unit stellar mass in the chosen band. If ψ is in $M_\odot \text{yr}^{-1} \text{Mpc}^{-3}$, it must be converted to $M_\odot \text{s}^{-1} \text{Mpc}^{-3}$ before multiplying by a yield in photons $M_\odot^{-1}$.

For dust emission, the correct source is not an independent stellar term but the absorbed stellar and AGN energy re-emitted in the infrared. If $\dot{u}_{\text{abs}}$ is an absorbed energy rate density and $\bar{E}_{\text{dust}}$ is an average infrared photon energy, then a rough photon source rate is

$$\dot{N}_{\gamma, \text{dust}} \sim \frac{\dot{u}_{\text{abs}}}{\bar{E}_{\text{dust}}}. \quad (\text{D25})$$

This term must be tied to the absorbed part of the stellar or AGN emissivity so that the absorbed photons are not also counted as escaped direct photons.

Schematic terms for supernovae, cosmic-ray interactions, and reionization can be useful in a broad source inventory, but they are not leading terms in the present optical–infrared EBL photon count. They should therefore remain explanatory unless a calculation explicitly integrates their spectra, redshift evolution, escape fractions, and absorption histories.

E. APPENDIX E. NUMERICAL CONVERSION CHECKS

The conversion from intensity to energy density is

$$u = \frac{4\pi}{c} I. \quad (\text{E26})$$

For

$$I = 1 \text{ nW m}^{-2} \text{ sr}^{-1} = 1 \times 10^{-9} \text{ W m}^{-2} \text{ sr}^{-1}, \quad (\text{E27})$$

this gives

$$u = \frac{4\pi}{2.998 \times 10^8 \text{ m s}^{-1}} (1 \times 10^{-9} \text{ W m}^{-2} \text{ sr}^{-1}) \simeq 4.19 \times 10^{-17} \text{ J m}^{-3}. \quad (\text{E28})$$

For photons of energy 1 eV = 1.602×10^{-19} J, this corresponds to

$$n_{\gamma} \simeq \frac{4.19 \times 10^{-17} \text{ J m}^{-3}}{1.602 \times 10^{-19} \text{ J}} \simeq 2.62 \times 10^2 \text{ m}^{-3} = 2.62 \times 10^{-4} \text{ cm}^{-3}. \quad (\text{E29})$$

For photons of energy 0.01 eV, the same intensity gives

$$n_{\gamma} \simeq 2.62 \times 10^{-2} \text{ cm}^{-3}. \quad (\text{E30})$$

This illustrates why the infrared and submillimeter EBL dominate photon number density more strongly than bolometric intensity.

For the FIRAS far-infrared background, a useful back-of-the-envelope check is

$$u_{\text{FIRB}} \simeq \frac{4\pi}{c} (14 \times 10^{-9}) \simeq 5.9 \times 10^{-16} \text{ J m}^{-3}. \quad (\text{E31})$$

If the representative photon energy is $\bar{E}_{\gamma} \sim 4\text{--}5 \times 10^{-3}$ eV, or $(6.4\text{--}8.0) \times 10^{-22}$ J, then

$$n_{\gamma, \text{FIRB}, 0} \sim \frac{5.9 \times 10^{-16}}{(6.4\text{--}8.0) \times 10^{-22}} \text{ m}^{-3} \sim (0.7\text{--}0.9) \text{ cm}^{-3}. \quad (\text{E32})$$

This agrees with the more careful statement that the measured FIR/submillimeter component alone contributes roughly $0.7\text{--}0.8 \text{ cm}^{-3}$.

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